Jeremy Lien

jjlien@umich.edu | 248-346-6885
www.linkedin.com/in/jjlien

EDUCATION

University of Michigan, Ann Arbor, Michigan **August 2024 - May 2027**
BS Computer Science, Minor in Electrical Engineering

- Relevant Coursework: VLSI Design I, Computer Organization, Quantum Computing, Data Structures & Algorithms, Digital Integrated Circuits, Logic Design

EXPERIENCE

Full-Custom 16-Bit RISC Microprocessor **Ann Arbor, Michigan**
Project **January 2026 - Present**

- Led design and execution of a 16-bit RISC microprocessor datapath utilizing the TSMC 65nm PDK, attaining a operational frequency of 400 MHz
- Built smallest-in-class 16-word x 16-bit register file with master-slave flip-flops and transmission gate logic, sized specifically to drive heavy capacitive bus load
- Created a Carry-Select/Carry-Bypass ALU supporting signed arithmetic (ADD, SUB, AND, OR, XOR) and dynamic Process Status Register (PSR) flag generation for conditional branching

8-Bit Dual-Mode Ripple-Carry Adder **Ann Arbor, Michigan**
Personal Project **October 2025 - December 2025**

- Designed an 8-bit dual-mode ripple-carry adder in Cadence Virtuoso using dynamic logic, transmission-gate full adders, and custom multiplexer/register topologies, with logical effort analysis to improve speed and power
- Achieved a 2.67 GHz maximum clock frequency in performance-optimized dynamic-logic design with low threshold voltage transistors for critical paths, representing a 424% improvement over preliminary design
- Developed a power-efficient dynamic-logic variant, reducing total power consumption by 64% at target frequency, verified using custom vector file based testing for both addition and accumulation modes

LC-2K Pipeline Processor Simulator **Ann Arbor, Michigan**
Personal Project **October 2025 - December 2025**

- Constructed a cycle-accurate 5-stage pipeline simulator (IF, ID, EX, MEM, WB) in C to emulate the LC-2K instruction set architecture, managing internal CPU state transitions and memory operations
- Implemented hazard detection and resolution logic, incorporating load-use stalling and data forwarding paths to resolve read-after-write (RAW) data hazards and control dependencies

Four Function Sequential Calculator **Ann Arbor, Michigan**
Personal Project **October 2025 - November 2025**

- Planned and implemented a functional four-operation calculator (add, subtract, multiply, divide) on an Intel DE2-115 FPGA utilizing Verilog and RTL design principles, achieving stable operation at 50MHz leveraging synchronous design practices
- Constructed a custom control finite-state machine (FSM) coordinating an 11-bit signed-magnitude datapath, including multiplexers, register control, add/sub unit, Booth multiplier logic, and iterative division

University of Michigan Climate and Space Sciences Department **Ann Arbor, Michigan**
Undergraduate Student Researcher **August 2024 - May 2025**

- Engineered a desktop application with GUI (Tkinter, PIL, visby, Matplotlib) to visualize Martian surface topography (up to 18,575,000 square miles) leveraging voxel-based 3D rendering and processing
- Engineered a GPU-accelerated 3D visualization engine using PyTorch tensors and VisPy, enabling interactive exploration of Martian terrain and reducing rendering time by 75%

Taiwanese Student Association **Ann Arbor, Michigan**
Finance Chair **August 2024 - Present**

- Raised $5000+ in sponsorships and donations by running targeted outreach, created monthly newsletter program to maintain engagement and keep alumni connected with program updates and opportunities
- Managed finances for large-scale events with $3,500+ spend and 500+ attendees, coordinating purchase orders, vendor payments, and reimbursements in line with university policy

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Rust, Verilog, VHDL, HTML/CSS, JavaScript
Frameworks/Tooling: Cadence Virtuoso, Quartus, PyTorch, Matplotlib, Altium Designer, NumPy, pandas, ROS2, MATLAB